8B Congruence, Similarity and Quadrilateral Properties

Standard Worksheet

Name: _____ Class: _____ Date: _____

PART A · FLUENCY & SKILL PRACTICE

Show complete step-by-step working in your workbook

Q1. **State which congruence test (SSS, SAS, ASA, RHS) proves each pair of triangles is congruent:**

(a) Sides $5\ \text{cm}, 7\ \text{cm}, 9\ \text{cm}$ in both triangles

(b) Sides $6\ \text{cm}, 8\ \text{cm}$ and included angle 55° in both

(c) Angles $40^\circ, 75^\circ$ and included side $10\ \text{cm}$ in both

(d) Right angle 90°, hypotenuse $15\ \text{cm}$, leg $9\ \text{cm}$ in both

Q2. **Find the unknown angle x° in each quadrilateral:**

(a) General quadrilateral with angles $82^\circ, 96^\circ, 115^\circ, x^\circ$

(b) Parallelogram with acute angle 64° and obtuse angle x°

(c) Kite with non-equal angles 92° and 54°, and two equal angles x°

(d) Trapezium with co-interior angles 71° and x° between parallel sides

Q3. **Given $\triangle ABC \sim \triangle DEF$ with $AB = 4\ \text{cm}, BC = 6\ \text{cm}, AC = 7\ \text{cm}$, and $DE = 10\ \text{cm}$, find:**

(a) The scale factor k from $\triangle ABC$ to $\triangle DEF$

(b) The length of side EF

(c) The length of side DF

(d) The ratio of their perimeters $\text{Perimeter}(\triangle DEF) : \text{Perimeter}(\triangle ABC)$

Q4. **Identify the special quadrilateral described by each set of properties:**

(a) All 4 sides equal; diagonals bisect each other at 90° (angles not necessarily 90°)

(b) Opposite sides equal; all 4 interior angles 90°; diagonals equal in length

(c) Two distinct pairs of adjacent sides equal; one pair of opposite angles equal

(d) Exactly one pair of parallel sides

PART B · PROBLEM-SOLVING & APPLICATIONS

Write clear mathematical statements with units

Q5. A vertical $2.0\ \text{m}$ street sign in Bendigo casts a horizontal shadow of length $1.6\ \text{m}$. At the same time, a nearby heritage clock tower casts a shadow of length $18.4\ \text{m}$. Using similar triangles, find the height of the clock tower.

Q6. In a rectangle $ABCD$, the diagonals AC and BD intersect at M. Given that $AM = 8.5\ \text{cm}$ and side $AB = 8\ \text{cm}$, find (a) the full length of diagonal BD and (b) the length of side BC.

Q7. An A-frame roof truss is an isosceles triangle ABC with $AB = AC = 6.5\ \text{m}$ and horizontal span $BC = 12.0\ \text{m}$. A vertical king-post AM connects the apex A to the midpoint M of BC. State the congruence test that proves $\triangle ABM \cong \triangle ACM$ and calculate the height AM.

Q8. The four interior angles of a convex quadrilateral are $(x + 15)^\circ, (2x - 5)^\circ, (2x + 10)^\circ$, and $(3x - 20)^\circ$. Solve for x and find the four angles.

PART C · REASONING & CHALLENGE

Justify your conclusions using algebraic or geometric reasoning

Q9. Explain why the condition AAA (three pairs of equal angles) guarantees that two triangles are **similar**, but does NOT guarantee that they are **congruent**.

Q10. In rectangle $ABCD$, prove that the two diagonals AC and BD have equal length by showing that $\triangle ABC \cong \triangle BAD$ using the SAS congruence test.